Database Management Systems

A Database Management System (DBMS) is software that provides an organized mechanism for storing, managing, retrieving, and protecting data. A DBMS provides data definition, data manipulation, transaction management, concurrency control, security, backup, and recovery.

Database Normalization

Database normalization is a systematic technique for organizing relational database tables to reduce redundancy and improve data integrity. Without normalization, the same information may be stored repeatedly in multiple rows. Repetition can lead to update, insertion, and deletion anomalies.

First Normal Form

First Normal Form, or 1NF, requires every attribute to contain atomic values and removes repeating groups. A relation should have a well-defined structure where each cell contains a single value. First Normal Form provides the foundation for further normalization.

Second Normal Form

Second Normal Form, or 2NF, builds upon First Normal Form. A relation must already satisfy 1NF and every non-key attribute must depend on the entire primary key. This is especially important when a relation has a composite primary key.

Third Normal Form

Third Normal Form, or 3NF, builds upon Second Normal Form. A relation satisfies 3NF when it has no undesirable transitive dependencies between non-key attributes. Removing transitive dependencies helps reduce redundancy and improve consistency.

Database Indexing

Database indexing improves the speed of data retrieval. An index is an additional data structure that allows a database engine to locate records more efficiently. Common indexing structures include B-tree indexes and hash indexes.